

Module 2: $\text{kappa} = \text{phi}(143) * \text{c_lemma} / 1\text{e}10 = 4.8433014197780389$

Machine Certificate v1.6 | David Fox | May 21, 2026
Opera Numerorum -- Tendon B (80-bit long double, gcc)

1. Claim

The constant kappa is defined by Lemma 4.1 (conductor constant) as:

```
kappa := phi(143) * c_lemma / 1e10
        = 120 * 403608451.6483666 / 1e10
        = 4.8433014197780389
```

where **phi(143) = 120** (Euler totient of 143 = 11 x 13) and **c_lemma = 403608451.6483666** is the Lemma 4.1 conductor constant computed to 80-bit long double precision by the C binary print_kappa.

2. Source Code

File: **bin/print_kappa.c**

```
/* Battle Plan v1.6 - Module 2
 * kappa = phi(143) * c_lemma / 1e10
 * Uses 80-bit long double for c_lemma (gcc x86_64). */
#include <stdio.h>

int main(void) {
    long double c_lemma = 403608451.6483666L;
    long double phi_143 = 120.0L;
    long double kappa = phi_143 * c_lemma / 1e10L;
    printf("%.16Lg\n", kappa);
    return 0;
}
```

3. Build Environment

```
$ gcc --version | head -1
gcc (GCC) 12.x -- x86_64, 80-bit long double

$ gcc -O3 -std=c11 bin/print_kappa.c -o bin/print_kappa -lm

$ sha256sum bin/print_kappa.c
d9a638794b092f55c06f0ef099cf076f4bf85743b8e5e6c211ead4013640cf92  bin/print_kappa.c

$ sha256sum bin/print_kappa
5d8be2b770dd02cc1eb27eba784e402452474eddc2460919e2c1fbc7b5fbd22  bin/print_kappa

$ ./bin/print_kappa
4.8433014197780389
```

4. Raw Execution Log

```
$ ./bin/print_kappa
4.8433014197780389
```

5. Causal Dependency on Module 1

Module 2 uses **phi(143) = 120** which depends on the factorization $143 = 11 \times 13$, not on the numerical value of alpha_0 from Module 1. The causal parent SHA is recorded for chain integrity.

Module 1 parent SHA (stdout of alpha0.py):

63ef870a78766619327e99b68683bceff8c8ef9a525298756c77c8378fd2c291

6. Cryptographic Binding

Item		SHA-256 Digest
Source	bin/print_kappa.c	d9a638794b092f55c06f0ef099cf076f4bf85743b8e5e6c211ead4013640cf92
Binary	bin/print_kappa	5d8be2b770dd02cc1eb27eba784e402452474eddc2460919e2c1fbc7b5fbd22
Stdout	m2.out	3716c7dbb32524074b8fffb65eea45069c8b568a31dc73706405116b84029a83
Causal parent (M1 stdout)		63ef870a78766619327e99b68683bceff8c8ef9a525298756c77c8378fd2c291

7. Verification Commands

```
# Recompile and verify:
$ gcc -O3 -std=c11 bin/print_kappa.c -o bin/print_kappa -lm
$ sha256sum bin/print_kappa.c # expect d9a638794b092f55...
$ sha256sum bin/print_kappa   # expect 5d8be2b770dd02cc...

# Verify stdout:
$ ./bin/print_kappa | sha256sum
3716c7dbb32524074b8fffb65eea45069c8b568a31dc73706405116b84029a83 -

# Verify causal parent:
$ python3 certificates/alpha0.py | sha256sum
63ef870a78766619327e99b68683bceff8c8ef9a525298756c77c8378fd2c291 -
```

Reproduce: `./bin/print_kappa | sha256sum`

Repository: <https://github.com/DavidFox998/alpha0-ponti> -- Certificate generated by Machine Certificate v1.6